24/01/2026
Morning

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

Memory Based
Answers & Solutions

Time : 3 hrs. M.M. : 300

for

JEE (Main)-2026 (Online) Phase-1

(Physics, Chemistry and Mathematics)





PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. A spring of stiffness k= 15 N/m is cut into a ratio of 3 : 1. Find the spring constant of smaller length spring thus formed.

(1) 15 N/m (2) 30 N/m

(3) 45 N/m (4) 60 N/m

Answer (4)

Sol. $\frac{1}{k}=\frac{4}{k_1}$

$\Rightarrow k_1 = 4k$

$\Rightarrow k_1$ = 60 N/m

2. EM waves and their source are given

	Column-I		Column-II
(a)	X-rays	(p)	Hot bodies & Molecules
(b)	Infrared rays	(q)	Oscillating current in antennas
(c)	Microwaves	(r)	Magnetron
(d)	Radio waves	(s)	Fast moving electrons Striking a metal plate

(1) (a)-(p); (b)-(s); (c)-(r); (d)-(q)

(2) (a)-(s); (b)-(p); (c)- (r); (d)-(q)

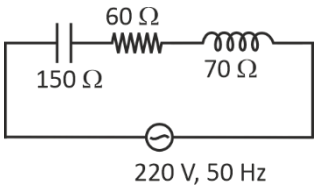
(3) (a)-(s); (b)-(p); (c)- (s); (d)-(q)

(4) (a)-(s); (b)-(r); (c)- (p); (d)-(q)

Answer (2)

Sol. Theoretical

3. For the given AC circuit find the power factor.

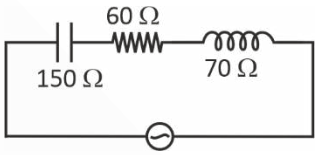


(1) $\frac{4}{5}$ (2) $\frac{3}{5}$

(3) $\frac{3}{4}$ (4) $\frac{4}{3}$

Answer (2)

Sol.



$X = |X_L - X_C| = 80\ \Omega$

$R = 60\ \Omega$

$\Rightarrow \tan\theta = \frac{80}{60} = \frac{4}{3}$

$\Rightarrow \cos\theta$ = Power factor = $\frac{3}{5}$

4. In H-like atom ratio of speed in two orbits is 3 : 2, then ratio of energy is

(1) 2 : 3 (2) 9 : 4

(3) 2 : 1 (4) 5 : 3

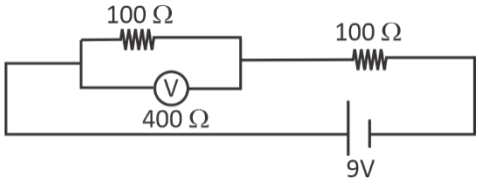
Answer (2)

Sol. $v = \frac{Z}{n}$

$KE = \left(\frac{Z}{n}\right) = 9:4$



5. A voltmeter of 400 Ω resistance is in parallel with 100 Ω resistor. And the combination is connected with 100 Ω resistor and a battery of 9 volt in series as shown. Find the reading of voltmeter.



(1) 5 volts (2) 3 volts

(3) 4 volts (4) 6 volts

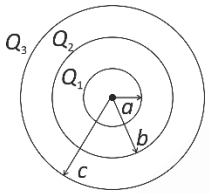
Answer (3)

Sol. $R_{eq.} = \frac{400\times100}{500} + 100 = 180$

$I = \frac{9}{180}$

So, V(voltmeter) = $\frac{9}{180}\times80$ = 4 volts.

6. Three concentric uniformly charged shells are kept as show. Find potential of the each shell.



(1) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$

$V_B = \frac{k(Q_1+Q_2+Q_3)}{b}$

$V_C = \frac{k(Q_1+Q_2+Q_3)}{c}$

(2) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$

$V_B = \frac{k(Q_1+Q_2)}{b} + \frac{kQ_3}{c}$

$V_C = \frac{k(Q_1+Q_2+Q_3)}{c}$

(3) $V_A = \frac{kQ_1}{a} + \frac{k(Q_2+Q_3)}{c}$

$V_B = \frac{k(Q_1+Q_2)}{b} + \frac{kQ_3}{c}$

$V_C = \frac{k(Q_1+Q_2+Q_3)}{c}$

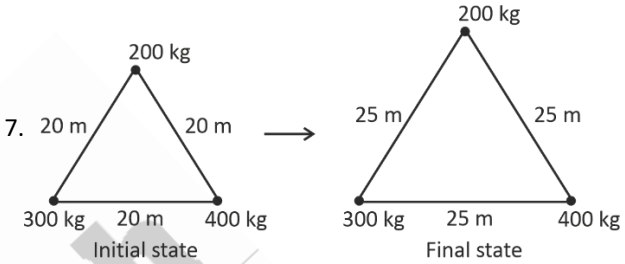
(4) $V_A = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$

$V_B = \frac{k(Q_1+Q_2)}{a} + \frac{kQ_2}{b}$

$V_C = \frac{k(Q_1+Q_2+Q_3)}{c}$

Answer (2)

Sol. Formula based

7. 

Find the work done.

(Given: $G = 6.67 \times 10^{-11}$ N-m^2/kg^2)

(1) 1.7342×10^{-7} J (2) 1.6253×10^{-7} J

(3) 2.5232×10^{-7} J (4) 6.6325×10^{-7} J

Answer (1)

Sol. $U_i = -\frac{G\times200\times300}{20} - \frac{G\times200\times400}{20} - \frac{G\times300\times400}{20}$

$U_i = -\frac{26\times10^4\times G}{20} = -8.671\times10^{-7}$ J

$U_f = -\frac{G\times200\times300}{25} - \frac{G\times200\times400}{25} - \frac{G\times300\times400}{25}$

$U_f = -\frac{26}{25}\times10^4\times6.67\times10^{-11}$

$U_f = -6.9368 \times 10^{-7}$ J

$W = \Delta U = 1.7342 \times 10^{-7}$ J



8. Match the two lists given below .

	List-I		List-II
a.	Magnetic flux	1.	$M^1L^2T^{-2}A^{-2}$
b.	Magnetic permeability	2.	$M^1L^2T^{-2}A^{-1}$
c.	Magnetic induction	3.	$M^1L^1T^{-2}A^{-2}$
d.	Self-induction	4.	$M^1L^0T^{-2}A^{-1}$

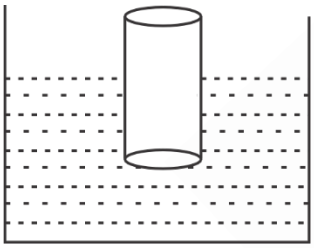
(1) a-2, b-3, c-4, d-1 (2) a-3, b-2, c-1, d-4

(3) a-4, b-3, c-1, d-2 (4) a-1, b-2, c-3, d-4

Answer (1)

Sol. Theoretical

9. A cylinder of mas *m*, length *l* and area of cross section *A* is in equilibrium in liquid of density ρ. Find time period of small vertical oscillations.



(1) $2\pi\sqrt{\frac{mA}{\rho g}}$ (2) $2\pi\sqrt{\frac{mg}{\rho A}}$

(3) $2\pi\sqrt{\frac{m}{\rho A^2 g}}$ (4) $2\pi\sqrt{\frac{m}{\rho A g}}$

Answer (4)

Sol. $F = (\rho Ag)x$

$$T = 2\pi\sqrt{\frac{m}{\rho Ag}}$$

10. A dipole is placed in uniform magnetic field B = 800 gauss at an angle 30° then it experiences the torque of 16×10^{-3} N-m. Find the work done in slowly moving the dipole from stable equilibrium to unstable equilibrium.

(1) 12.8×10^{-3} J (2) 5×10^{-3} J

(3) 24.5×10^{-3} J (4) 7.6×10^{-3} J

Answer (1)

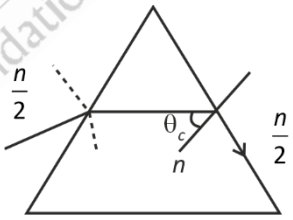
Sol. $\left|\vec{\tau}\right| = \left|\overrightarrow{M} \times \vec{B}\right| = \frac{\left|\overrightarrow{M} \times \vec{B}\right|}{2}$

$$\Rightarrow \left|\overrightarrow{M}\right| = \frac{2 \times 16 \times 10^{-3}}{800 \times 10^{-4}} = \frac{4}{10}$$

$$\text{So } \Delta W = 2\left|\overrightarrow{M}\right|\left|\vec{B}\right| = \frac{2 \times 4}{10} \times 800 \times 10^{-4}$$

$$\Rightarrow \Delta W = 64 \text{ mJ}$$

11. A light ray incident on the prism such that deviation is minimum and angle of incidence on 2nd surface is critical angle. Find prism angle.



(1) 90° (2) 60°

(3) 105° (4) 74°

Answer (2)

Sol. $\sin\theta_c = \frac{1}{2}$

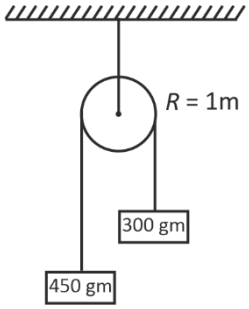
$\theta_c = 30°$

$r_1 = r_2 = 30°$

$A = 30 + 30 = 60°$



12. When system is released from rest the heavier mass goes 81 cm in 9 sec, find rotational inertia. (g = 10 m/s^2)



(1) 74.25 kg-m^2

(2) 100.25 kg-m^2

(3) 50.25 kg-m^2

(4) 25.25 kg-m^2

Answer (1)

Sol. $s=\frac{1}{2}at^2$

$0.81=\frac{1}{2}a\times 9^2$

a = 0.02 m/s^2

$a=\frac{(0.45-0.3)\times 10}{0.45+0.3+\frac{I}{R^2}}$

$0.02=\frac{0.15\times 10}{0.75+\frac{I}{R^2}}$

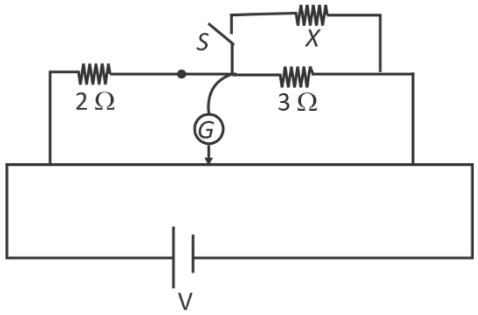
$0.75+\frac{I}{R^2}=75$

$\frac{I}{R^2}=74.25$

I = 74.25 × 1^2

I = 74.25 kg-m^2

13. In meter bridge diagram given below, if S is closed, null point shifts by 25 cm. Find value of resistance X. (in cm).



(1) 4.1 (2) 1.68

(3) 6.28 (4) 5.4

Answer (2)

Sol. $\Rightarrow \frac{2}{3}=\frac{l}{100-l} \Rightarrow l$ = 40 cm

Now $\frac{2(3+x)}{3x}=\frac{65}{35}=\frac{13}{7}$

$\Rightarrow$ 42 + 14x = 39x

$\Rightarrow$ 42 = 25x $\Rightarrow$ x = 1.68 Ω

14. **Statement-1 :** Binding energy per nucleon always increase with mass number.

Statement-2 : Binding energy per nucleon for smaller mass number always performs nucleon fusion.

(1) Statement-1, true Statement-2, false

(2) Statement-1, true Statement-2, true

(3) Statement-1, false Statement-2, true

(4) Statement-1, false Statement-2, false

Answer (3)

Sol. Statement-1 False → Binding energy per nucleons does net always increase with mass number.

Statement-2 : True → Light nuclei with low binding energy per nucleon generally undergo nucleon fusion to become stable by increasing their binding energy per nucleons towards the iron-56 peak.



15. Velocity of electron in n^{th} shell of a hydrogen like atom is 3×10^5 m/s and velocity of electron in m^{th} shell of that atom is 2.5×10^5 m/s. Find ratio of radius of mth shell to n^{th} shell.

(1) $\frac{25}{40}$ (2) $\frac{25}{36}$

(3) $\frac{36}{25}$ (4) $\frac{36}{35}$

Answer (3)

Sol. $V \propto \frac{z}{n}$ and $r \propto \frac{n^2}{z}$

$\frac{V_n}{V_m} = \frac{m}{n} = \frac{6}{5}$

$\frac{r_m}{r_n} = \frac{m^2}{n^2} = \frac{36}{25}$

16.

17.

18.

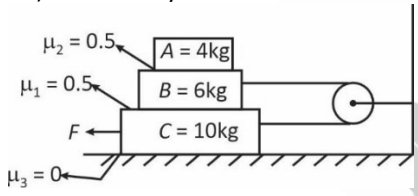
19.

20.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. For the given arrangement find the value of F(in Newton) so that body c moves with constant velocity.



Answer (100)

Sol. $fr_{(max)} = \frac{1}{2} \times 10 \times 10 = 50$ N

Since B also moves with uniform velocity

So, friction between A & B is zero.

Now, $F - T - 50 = 0$

$\Rightarrow F = 50 + T$ ....(1)

Also, $T - 50 = 0$ $T = 50$ N

So, $F = 100$ N

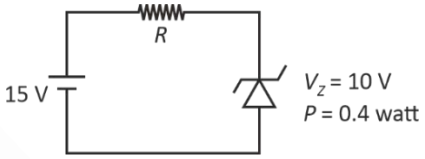
22. For a microscope focal length of objective is 2 cm and focal length of eyepiece is 4 cm. Tube length is L = 10 cm. Magnification for normal adjustment is 5^x. Find the value of x.

Answer (2)

Sol. $M = \frac{1}{fo} \cdot \frac{D}{fe}$

$\mathbf{M} = \frac{10}{2} \cdot \frac{25}{5} = 25 = (5)^2$

23. For the given circuit the breakdown voltage of Zener diode is $V_Z = 10$ volts and it can with-stand the power dissipation of 0.4 watt. Find the value of resistance R (in Ω)



Answer (125)

Sol. Current in Zener diode is

$I_{(Z)} = \frac{4}{10 \times 10} = \frac{4}{100}$ Ampere

So, $R = \frac{(15-10)}{4} \times 100 = 125\,\Omega$

24. If potential varies as distance r as $v(r) = ar^3 + b$. Total magnitude of charge Q inclosed within a sphere of unit radius is $Q = \alpha(\pi a \varepsilon_0)$. Find the value of α.

Answer (12)

Sol. $v(r) = ar^3 + b$

$\Rightarrow E(r) = \frac{-dv}{dr} = -3ar^2$

So $\int E \cdot ds = \frac{|q_{in}|}{\varepsilon_0}$

$\Rightarrow 3ar^2 \cdot 4\pi r^2 = \frac{|q_{in}|}{\varepsilon_0}$

$\Rightarrow |q_{in}| = 12\pi a \varepsilon_0 (1)^4 = 12\pi a \varepsilon_0$

So, $\alpha = 12$

25



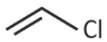
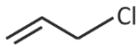
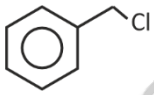
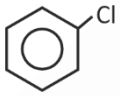
CHEMISTRY

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

1. Match List-I with List-II.

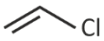
List-I	List-II
A. Vinyl halide	(I) 
B. Allyl halide	(II) 
C. Benzyl halide	(III) 
D. Aryl halide	(IV) 

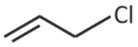
Select the correct option.

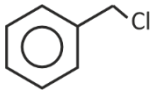
(1) A(II), B(I), C(III), D(IV) (2) A(I), B(II), C(III), D(IV)

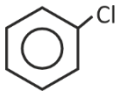
(3) A(I), B(II), C(IV), D(III) (4) A(II), B(I), C(IV), D(III)

Answer (2)

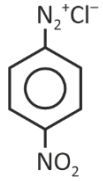
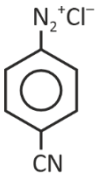
Sol. Vinyl halide → 

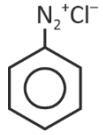
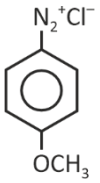
Allyl halide → 

Benzyl halide → 

Aryl halide → 

2. The correct order of stability of following diazonium ions is

a.  b. 

c.  d. 

(1) a < b < c < d (2) a < b < d < c

(3) c < d < b < a (4) d < c < b < a

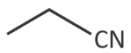
Answer (1)

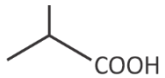
Sol. Stronger the electron withdrawing group attached at para position of $-N_2^+$ in diazonium ion, lesser is the stability and more electrophilicity.

Stability : (d) > (c) > (b) > (a)

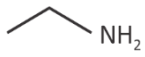
3. $A \xrightarrow[\text{II. } H_3O^+]{\text{I. } SnCl_2/HCl} B \xrightarrow[H^+]{KMnO_4} C \xrightarrow[H_2O]{Cl_2, P} C_3H_5O_2Cl$

Final product has one chiral centre. Structure of A is

(1) 

(2) 

(3) 

(4) 

Answer (1)

